DE LA SALLE UNIVERSITY
College of Science
Department of Physics

SEMPY01: Semiconductor Physics
3 units
Pre-requisite: SOLST01
Prerequisite To:

Instructor:

Consultation Hour:

Contact details:
Class Schedule and Room:

Course Description

The course deals with the physics of semiconductors and semiconductor devices. It includes behavior of electrons moving in a periodic potential and the features of the energy band structure that distinguish semiconductors from metals and insulators; the statistics of electrons and holes in semiconductors in equilibrium, behavior of excess carriers in semiconductors; properties of a pn junction, the transistor, LEDs, semiconductor lasers, and other semiconductor devices.

Learning Outcomes

On completion of this course, the student is expected to present the following learning outcomes in line with the Expected Lasallian Graduate Attributes (ELGA)

ELGA	Learning Outcome
Critical and Creative Thinker	LO1. Infer and predict relevant consequences and apply the laws and principles in solving semiconductor physics problems.
Effective Communicator	LO2. Interpret and explain the solutions in a clear and organized manner.
Lifelong Learner Service-Driven Citizen	LO3. Apply the basic physical laws and principles of semiconductor physics in the relation to its environment and everyday living.

Final Course Output

As evidence of attaining the above learning outcomes, the student is required to perform and submit the following during the indicated dates (*subject to change due to holidays/cancellation of class due to natural calamity/and other unforeseen events) of the term.

Learning Outcome	Required Output	Due Date
LO1. Infer and predict relevant consequences and apply the laws and principles in solving physics problems.	HW, Assignments, Practice Problem, Problem Sets	
	Long Exams	Week 4, 8, 11
LO2. Interpret and explain the solutions in a clear and organized manner.	Concept discussion, essay, reflection paper	
LO3. Apply the basic physical laws and principles of physics in the relation to its environment and everyday living.		
	Final Examination	Week 14

Rubric for assessment HW, Assignments, Practice Problem, Problem Sets Concept discussion, essay, reflection paper				
CRITERIA	**Excellent (4)**	**Good (3)**	**Satisfactory (2)**	**Needs Improvement (1)**
Understanding and Knowledge 50%	Shows deep understanding of the problem, the physics and mathematical concepts and processes needed to solve the problem.	Shows substantial understanding of the problem, the physics and mathematical concepts and processes needed to solve the problem.	The solution is incomplete. Only some parts of the problem are understood.	There is no solution provided or the solution is not related/appropriate to the problem.
Explanation 30%	Complete presentation of solution with detailed explanation.	Good presentation with clear explanation. Arguments presented have some flaws.	Explanation is incomplete or unclear.	No explanation provided or the explanation is not related to the problem.
Integration & Synthesis 20%	Demonstrates integration and synthesis of several physics concepts.	Demonstrates some integration and synthesis of concepts.	Demonstrates limited integration and synthesis of concepts.	No integration and synthesis of concepts demonstrated.

Additional Requirements
Review paper/s, attendance in webinar/s, conference, e-poster, Oral presentation of problem sets, Video Solutions, Physics Fieldtrip, Alternative Classes, Library Work, Physics Seatwork, Outreach Project, Community Volunteer, and other requirements prescribed in the course

Grading System

		Scale:	
3 Long Exams	60%	94-100	4.0
Portfolio of Problem Sets and Submission		87-93	3.5
and presentation of Final Product	25%	80-86	3.0
Teachers Evaluation:	15%	73-79	2.5
(*covers the overall performance based on required output)		66-72	2.0
		58-65	1.5
TOTAL:	**100%**	50-57	1.0
Passing Grade:	**50%**	<50	0.0

Learning Plan

LEARNING OUTCOME	**TOPIC**	**WEEK NO.**	**LEARNING ACTIVITIES**
LO1, LO2, LO3, LO4	Orientation	1	**Discussion of policies and grade requirements**
	Chapter 1 Crystal Structure of Solids	2-3	Chapter 1 1.1 Semiconductor Materials 1.2 Types of Solids 1.3 Space Lattices 1.3.1 Primitive and Unit Cell 1.3.2 Basic Crystal Structures 1.3.3 Crystal Planes and Miller Indices 1.3.4 Directions in Crystals 1.4 The Diamond Structure 1.5 Atomic Bonding *1.6 Imperfections and Impurities in Solids 1.6.1 Imperfections in Solids 1.6.2 Impurities in Solids *1.7 Growth of Semiconductor Materials 1.7.1 Growth from a Melt 1.7.2 Epitaxial Growth
	Chapter 2	4-5	2.1 Principles of Quantum Mechanics

	Introduction to Quantum Mechanics		2.1.1 Energy Quanta 2.1.2 Wave–Particle Duality 2.1.3 The Uncertainty Principle 2.2 Schrodinger's Wave Equation 2.2.1 The Wave Equation 2.2.2 Physical Meaning of the Wave Function 2.2.3 Boundary Conditions 2.3 Applications of Schrodinger's Wave Equation 2.3.1 Electron in Free Space 35 2.3.2 The Infinite Potential Well 2.3.3 The Step Potential Function 2.3.4 The Potential Barrier and Tunneling 2.4 Extensions of the Wave Theory to Atoms 2.4.1 The One-Electron Atom 2.4.2 The Periodic Table
			First Long Examination
LO1, LO2, LO3, LO4	Introduction to the Quantum Theory of Solids	6	Formation of energy bands Kronig-Penney model Electron effective mass Concept of holes Metals, insulators, semiconductors 4th Hour Activity: Exercise Problems
LO1, LO2, LO3, LO4	Introduction to the Quantum Theory of Solids	7	Density of states function Fermi-Dirac probability function Distribution function and the Fermi Energy 4th Hour Activity: Exercise Problems
	Semiconductors in Equilibrium	8	Equilibrium distribution of electron and holes Intrinsic semiconductors Extrinsic semiconductors Statistics of donors and acceptors 4th Hour Activity: Exercise Problems
			Second Long Examination
LO1	Carrier Transport Phenomena	9	Carrier drift Carrier diffusion Mobility Diffusion coefficient Graded impurity distribution
	Nonequilibrium Excess Carriers in Semiconductors	10	Carrier generation and recombination Mathematical analysis of excess carriers Ambipolar transport Quasi Fermi levels Excess carrier lifetimes 4th Hour Activity: Exercise Problems
LO5	The pn Junction	11	Basic structure Zero applied bias Reverse applied bias Forward bias I-V characteristics
	Metal-Semiconductor Contacts		Potential barrier Schottky barriers Ohmic contacts 4th Hour Activity: Exercise Problems
LO7, LO8	Semiconductor Devices	12	Optoelectronic devices Semiconductor power devices

			4th Hour Activity: Exercise Problems
LO13, LO14	Final Product	13	Portfolio of Problem Sets Submission and presentation of Final Product
	Final Exams	14	

References

1. Kittel. *Introduction to Solid State Physics*, 8th Ed, 2004, John Wiley & Sons.
2. Neamen, D. *Semiconductor Physics and Devices: Basic Principles*, Latest Edition, McGraw-Hill.
3. Neamen, D. *An Introduction to Semiconductor Devices,* Latest Edition, McGraw-Hill.
4. McGowan, Kevin Semiconductors : From Book To Breadboard CENGAGE
5. Sze. *Physics of Semiconductor Devices*, Latest edition, Wiley International.
6. Pierret, R. *Advanced Semiconductor Fundamentals vol.VI*, Modular Series on Solid State Devices. Purdue University.
7. Pierret, R. *Semiconductor Device Fundamentals.* Addison-Wesley Publishing Company
8. Van Zeghbroeck, B. *Principles of Semiconductor Devices*, Latest edition. University of Colorado Website http://ece-www.colorado.edu

Online Resources

AK Lectures. 2021. *Solid State Physics*. https://aklectures.com/subject/modern-physics/chemical-bonds-and-solid-state-physics University of Cambridge, 2020.

Introduction to Semicondutors. https://www.doitpoms.ac.uk/tlplib/semiconductors/index.php

Markvart and Castaner. 2012. Chapter IA-2: Semiconductor Materials and Modelling.

Practical Handbook on Photovoltaics, Second ed.Elsevier. Accessible: https://www-sciencedirect-com.dlsu.idm.oclc.org/science/article/pii/B9780123859341000027#s0015

Class Policies

- This is a major course in physics for physics students of the College of Science.
- Students are encouraged to get the latest edition of the textbook based on the references as required by the Instructor.
- It is the responsibility of the student to be mindful of his/her absences/tardiness and performance in class.
- Specific class policies on attendance, make-up quizzes, academic honesty, grading system, dress code, classroom management, and others not mentioned above, are to be discussed by the faculty member on the first day of classes.

Approved by:

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